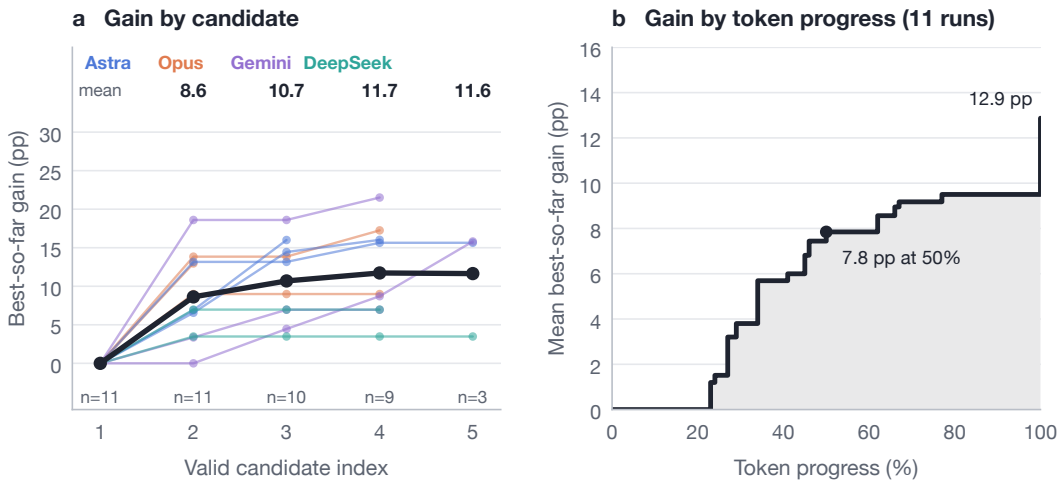
a Gain by candidate
Astra
Opus
Gemini
DeepSeek
mean
8.6
10.7
11.7
11.6
Best-so-far gain (pp)
30
25
20
15
10
5
0
n=11
n=11
n=10
n=9
n=3
1
2
3
4
5
Valid candidate index
b Gain by token progress (11 runs)
Mean best-so-far gain (pp)
16
14
12
10
8
6
4
2
0
12.9 pp
7.8 pp at 50%
0
20
40
60
80
100
Token progress (%)